# COURTNEY CARREIRA

(she/her)

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## EDUCATION

**Ph.D. Student** September 2023 - Present
Department of Astronomy & Astrophysics, The University of California, Santa Cruz *Santa Cruz, CA*
Advisor: Professor Brant Robertson

**M.S. Astronomy and Astrophysics** June 2025
Department of Astronomy & Astrophysics, The University of California, Santa Cruz *Santa Cruz, CA*

**B.S. Physics** Graduated with General Honors, May 2023
Department of Physics and Astronomy, Johns Hopkins University *Baltimore, MD*
Minor in Applied Mathematics and Statistics

## RESEARCH EXPERIENCE

**Graduate Student Researcher** January 2024 - Present
Department of Astronomy & Astrophysics, The University of California, Santa Cruz *Santa Cruz, CA*

- As a member of the *JWST* Advanced Deep Extragalactic Survey (JADES) collaboration, I am working to understand how morphology traces galaxy evolution across cosmic time. I organize the Morphology Working Group within the collaboration.
- Created catalogs of morphological parameters by using Bayesian techniques to robustly perform model-fitting of Sérsic profiles to $> 10^5$ galaxies in GOODS-N and GOODS-S, resulting in over 3 million Sérsic profiles computed and shared with the community as part of JADES Data Release 5.

**NSF REU Intern and NRAO SOS Researcher** June 2022 - September 2023
Smithsonian Astrophysical Observatory *Cambridge, MA*

- Using observations of atomic and molecular gas emissions in M33 to analyze the effect of proximity from the southeastern spiral arm in the formation of molecular clouds; advised by Dr. Eric Koch and Dr. Sarah Jeffreson, within Professor Alyssa Goodman's research group.
- Ongoing work resulted in a successful NRAO Student Observing Support award to obtain observations that resolve the filamentary morphology of molecular clouds across M33. Co-PI: Eric Koch, Title: *Linking the Resolved Filamentary Molecular ISM to Massive Star Formation across M33*.

**Undergraduate Researcher** May 2021 - May 2022
Department of Physics and Astronomy, Johns Hopkins University *Baltimore, MD*

- Collected photometric and spectroscopic data for a large set of low-metallicity stellar objects, believed to host transiting exoplanets; advised by Professor Kevin Schlaufman.
- Utilized Python coding and packages to numerically analyze the stellar objects of interest.

**Undergraduate Research Intern** May 2021 - January 2022
The Johns Hopkins University Applied Physics Laboratory *Laurel, MD*

- Performed correlation analysis of simulated gamma-ray and UVOIR emissions from Type Ia supernovae, and assisted with scientific validation for mission proposal; advised by Dr. Richard S. Miller.

**CIRCUIT Intern** April 2020 - May 2021
The Johns Hopkins University Applied Physics Laboratory *Laurel, MD*

- Analyzed Monte Carlo simulations of volatile transport across the lunar surface, specifically looking at water and carbon dioxide; advised by Dr. Parvathy Prem and others.

## PUBLICATIONS

### First Author

1. Carreira, C., et al. (2026). JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Catalogs of inferred morphological properties of galaxies from JWST/NIRCam imaging in GOODS-N and GOODS-S. arXiv e-prints, arXiv:2601.15957.

### Co-Author

9. Rinaldi, P., et al. incl. Carreira, C. (2026). The Way We Tally Becomes the Tale: the Impact of Selection Strategies on the Inferred Evolution of Little Red Dots Across Cosmic Time. arXiv e-prints, arXiv:2604.07138.

8. Zhu, Y., et al. incl. Carreira, C. (2026). Clump-like Structures in High-Redshift Galaxies: Mass Scaling and Radial Trends from JADES. The Astrophysical Journal, 1000(2), 303. DOI: 10.3847/1538-4357/ae50f1

7. Cameron, A., Carreira, C., et al. (2026). JADES: Evolution of nitrogen abundances in star-forming galaxies from $z \sim 1.5 - 7$. arXiv e-prints, arXiv:2601.15964.

6. Hainline, K., et al. incl. Carreira, C. (2026). JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Photometrically Selected Galaxy Candidates at $z > 8$. arXiv e-prints, arXiv:2601.15959.

5. Robertson, B., et al. incl. Carreira, C. (2026). JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Photometric Catalog. arXiv e-prints, arXiv:2601.15956.

4. Johnson, B., et al. incl. Carreira, C. (2026). JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: NIRCam Imaging in GOODS-S and GOODS-N. arXiv e-prints, arXiv:2601.15954.

3. Danhaive, A., et al. incl. Carreira, C. (2025). Beyond the stars: Linking H$\alpha$ sizes, kinematics, and star formation in galaxies at $z \approx 4 - 6$ with JWST grism surveys and `geko`. MNRAS, 547(4), stag437. DOI: 10.1093/mnras/stag437

2. Shields, T., et al. incl. Carreira, C. (2025). JADES: low surface brightness galaxies at $0.4 < z < 0.8$ in GOODS-S. MNRAS, 546(4), stag202. DOI: 10.1093/mnras/stag202

1. Robertson, B., et al. incl. Carreira, C. (2024). Earliest Galaxies in the JADES Origins Field: Luminosity Function and Cosmic Star Formation Rate Density 300 Myr after the Big Bang. ApJ, 970(1), 31. DOI: 10.3847/1538-4357/ad463d

## PRESENTATIONS

### Oral Presentations

Characterizing the morphological evolution of high-redshift galaxies with JADES — April 2026
*BeyondHubble: Revisiting the Hubble Sequence Across Cosmic Time* — *KICC, Cambridge, UK*

Science Review: Galaxy Morphology & Kinematics — June 2025
*JADES Team Meeting, Madrid/Boston 2025* — *Cambridge, MA*

Revealing the relationship between galaxy formation and morphology across cosmic time with JADES — May 2025
*UCSC Friday Lunch Astrophysics Seminar Hour (FLASH)* — *Santa Cruz, CA*

Do spiral arms form molecular clouds? — August 2022
*SAO Astronomy REU Summer Symposium* — *Cambridge, MA*

### Poster Presentations

The Effect of Spiral Arms on Molecular Cloud Formation in M33 — January 2023
*241$^{st}$ Meeting of the American Astronomical Society* — *Seattle, WA*

Lunar Crater Maturity Analysis in Python: Developing a Toolkit for Ejecta Analysis — June 2021
*5$^{th}$ Planetary Data Workshop & Planetary Science Informatics & Analytics* — *Virtual*

Lunar Crater Maturity Analysis in Python: Developing a Toolkit for Ejecta Analysis — March 2021
*52$^{nd}$ Lunar and Planetary Science Conference* — *Virtual*

The Effect of Isotopic Composition and Surface Residence Times on Lunar Volatile Transport — March 2021
*52$^{nd}$ Lunar and Planetary Science Conference* — *Virtual*

## AWARDS AND RECOGNITION

**ARCS Foundation Scholar, Northern California Chapter** — May 2026
Achievement Rewards for College Scientists (ARCS) Foundation — *San Francisco, CA*

**Whitford Prize** — June 2025
Department of Astronomy & Astrophysics, The University of California, Santa Cruz — *Santa Cruz, CA*

- Awarded to the second-year UCSC Astronomy & Astrophysics graduate student who, in the judgment of the faculty, attains the highest achievement in research, coursework, and teaching.

**Graduate Research Fellowship Program (GRFP) Honorable Mention** — April 2025
National Science Foundation — *Alexandria, VA*

## TEACHING

**Teaching Assistant for ASTR 2** — January 2024 - March 2024
Department of Astronomy & Astrophysics, The University of California, Santa Cruz — *Santa Cruz, CA*

- Led one recitation section per week, which included a short lecture, group activities, and live demonstrations.
- Hosted office hours on a weekly basis, in collaboration with other TAs.

**Teaching Assistant for General Physics I** — August 2022 - December 2022
Department of Physics and Astronomy, Johns Hopkins University — *Baltimore, MD*

- During Active Learning sections of this course, worked closely with students as they completed a series of problems and hands-on demonstrations during their lectures.
- Hosted office hours on a weekly basis.

## OUTREACH AND SERVICE

**Lead Organizer, Department Journal Club** — January 2025 - Present
Department of Astronomy & Astrophysics, The University of California, Santa Cruz — *Santa Cruz, CA*

**PI, Osterbrock Rising Graduate Award Program** — March 2024 - Present
Osterbrock Leadership Program at the University of California, Santa Cruz — *Santa Cruz, CA*

- Created award program, in collaboration with the UCSC Women in Physics and Astrophysics organization, to provide $500 to four UCSC women and gender minority undergraduates to offset costs associated with applying to graduate school programs in physics and/or astronomy.
- Created additional award program to provide $500 to four UCSC undergraduates who are negatively impacted by the current immigration landscape (for example, students who are undocumented or DACAmented, or whose families are) to offset costs associated with applying to graduate school programs in STEM.
- Provided mentorship and support in developing application materials to the students selected to receive this award.
- Funding generously provided by the Osterbrock Leadership Program during the 2024 and 2025 Mini-Grant cycles.

**Local Organizing Committee** — January 2025
JADES Team Meeting, Santa Cruz 2025 — *Santa Cruz, CA*

*Updated as of May 18, 2026.*